

Answers to Practice Papers — 5

1. (b) One teacher can have many classes
2. (c) 153.67
3. (b) Update
4. (a) World Wide Web (WWW)
5. (b) Modem
6. (b) Underflow
7. (c) load()
8. (b) XML
9. (c) Alternate key
10. (c) Fibre Optic
11. (b) A linear data structure that follows a last-in-first-out (LIFO) order
12. (a) read() reads the entire file while readline() reads one line of the file
13. (d) The function is called, the function's statements are executed in order, and then the function returns a value
14. (a) It reads data from a binary file
15. (a) 5
16. (a) 6
17. (a) Both A and R true and R is the correct explanation for A.
18. (a) Both A and R true and R is the correct explanation for A.
19. **Correct code:-**

```
x= int(input("Enter value of x:"))
for in range (0,10):
    if x==y:
        print( x+y)
    else:
        print (x-y)
```

20. LAN is confined to one or nearby building but Internet has no specific geographical area. Internet is the collection of different LANs. It is the example of WAN.

OR

Advantages of Star Topology:

- (i) **Ease of service:** The star topology has a number of concentration point i.e. at the central node or at intermediate wiring closets. These provide easy occurs for service or re-configuration of the network
- (ii) **One device per connection:** In star topology, failure of a single connection typically involves disconnecting one node from an other.

Disadvantages of Star Topology:

- (i) **Long cable length:** Because each node in star topology is directly connected to the center, the star topology requires a large quantity of cable.
- (ii) **Central node dependency:** If the central node fails the network get failed.

21. **fetchall():** It fetches all the rows in a result set. If some rows have already been extracted from the result set, then it retrieves the remaining rows from the result set.

For example: cur.execute("Select * from Tablename")

Variable=cur.fetchall()

fetchmany(n): This method returns blocks of results according to a set limit. This method requires the desired number of records to be passed as an argument.

For example: cur.execute("Select * from Tablename")

Variable=cur.fetchmany(10) # fetch first 10 records

22. Candidate Key	Alternate Key
• A key that can be set as Primary key is called a candidate key.	• A key that can act as a primary key but is not selected as primary key
• The minimal set of attributes that can uniquely identify a tuple is known as a candidate key. For Example, STUD_NO in STUDENT relation.	• The candidate key other than the primary key is called an alternate key.

23. (a) bps – bit per second and NSFNET – National Science Foundation Network

(b) Packet switching is a method of transmitting data on a network by dividing it into small, manageable packets, and sending each packet individually, to be reassembled at its destination. It is used to allow multiple transmissions to occur simultaneously on a network, improving overall efficiency and reliability.

24. InIPpTtrRRW

25. **ALTER TABLE** is used to modify the structure of an existing table in a database. It allows you to add, modify, or delete columns, add or drop constraints, or make other changes to the table's structure.

`ALTER TABLE karmchari ADD COLUMN address VARCHAR(200);`

UPDATE is used to modify the data stored in a table. It allows you to change values in specific columns for one or more rows in the table.

`UPDATE karmchari SET salary = 80000.00 WHERE karmchari_id = 3;`

26. (a)

khachchar_id	naam	address	order_id	product
1	Ram Sharma	123 Main St	1	iPhone
1	Ram Sharma	123 Main St	3	iPad
2	Seema Patel	456 Elm St	2	MacBook
3	Ravi Singh	789 Oak Ave	4	Apple Watch

(b) (i)

avg(salary)
70000.000000

(ii)

pehla_naam	vibhag
Rashmi	Human Resources
Sachin	Sales

(iii)

aakhri_naam	pad
Sharma	Software Engineer
Verma	Sales Executive
Gupta	Operations Manager
Shukla	Customer Service Manager

(iv)

count(*)	max(salary)
9	95000.00

```
27. def JTOI():
    f=open("WORD.txt","r")
    data=f.read()
    for c in data:
        if c=='j':
            print('I', end="")
        else:
            print(c,end="")
    f.close()
```

OR

```
'''Adding multiple lines to the file'''
f=open("multiline.txt","a")
ch=True
while True:
    str=input("Enter line")
    f.write(str)
    f.write("\n")
    ch=input("Want to add more line Y/N")
    if ch=='N':
        break
print("Lines added to the file")
f.close()
```

28. (i) Select supcode, sum(price) from products group by supcode;

SUPCODE	SUM(PRICE)
SO1	13100
SO2	50000
SO3	12000

(ii) Select price * qty as amount from products where pid=102:

AMOUNT
55000

(iii) Select pname,sname from products natural join suppliers where qty>100;

PNAME	SNAME
DIGITAL CAMERA 14X	GET ALL INC
PEN DRIVE 16 GB	GET ALL INC

```
29. def SHIFT(lst):
    n=len(lst)
    print(lst)
    for i in range(n-1):
        for j in range(n-i-1):
            if lst[j]<0:
                lst[j ], lst[j+1 ]=lst [j+1], lst[j]
    print("List after shifting:",lst)
```

```

30. def PUSH(Books,N):
    for i in range(N):
        book id=int(input("Enter the book Id:"))
        bookName=input("Enter the book name:")
        bookCost=float(input( "Enter the cost:"))
        B=[Book-id, Bookname, book cost]
        Books append(b)
        if len (Books )==0:
            print("Empty stack")
        else: print(Books)

```

OR

```

def DISPLAY (Books, N):
    if books ==[]:
        print("Underflow!! Stack is empty")
    else:
        print("Book details")
        for i in range(len(Books),-1,-1):
            if book[3]<500:
                print(i)

```

31. (a) Star topology
 (b) Broadband
 (c) Switch/Hub
 (d) Radio waves
 (e) BLOCK B1

32. (a) Processed string: rACHNA*sAGAR*3134
 (b) Statement 1 : cursor = conn.cursor()
 Statement 2 : "SELECT * FROM BOOK WHERE Price BETWEEN 300 AND 500"
 Statement 3 : cursor.fetchall()

OR

- (a) Processed string: sqp*PAPER*6
 (b) Statement 1: mysql.connector.connect
 Statement 2: "SELECT * FROM BOOK WHERE bookname like 'A%' "
 Statment 3: cursor.fetchall()

```

33. def BUMPERf):
    import pickle
    f=open("GIFTS.DAT ","rb")
    try:
        print("Gifts remarks as "ON DISCOUNT")
        while True:
            x=pickle.load(f)
            if x["Remarks "]=="ONDISCOUNT":
                print(x)
    except EOFBrTor:
        pass
    f. close()

```

OR

```
def transfer():
    import pickle
    fr=open( "GIFTS.DAT ", "rb")
    fc=open("G COPY .DAT", "wb")
    try:
        while True:
            x=pickle.load(fr)
            if x["Price"]==500:
                pickle.dump (x,fc)
                print(x)
    except EOFError:
        pass
    fr.close()
    fc.close()
```

34. (a) Select * from consumer where address=='Delhi';
(b) Select * from Stationery where price between 8 AND 15;
(c) Select ConsumerName Address Company, Price from Consumer Stationery where Consumer.S ID = Stationery. SID;

OR

(c) Update Stationery set Price = Price +2;

35. (i) pickle
(ii) open("blocked.dat" , "wb") ; open("unblocked.dat","wb")
(iii) pickle. load(fin)
(iv) pickle.dump(rec,fblock);pickle.dump(rec, funblock)